```
import os
os.environ["OPENAI_API_KEY"] = "sk-proj-aOrFC5Kv6j6RomRGR-ACpT3BlbkFJ_oZ1CAo2shNwf4dwJK1rVFosrrfAyrH7
```

```
!pip install langchain chromadb faiss-cpu openai tiktoken langchain_openai langchain-community wikiped
```

```
Requirement already satisfied: langchain in /usr/local/lib/python3.11/dist-packages (0.3.23)
Requirement already satisfied: chromadb in /usr/local/lib/python3.11/dist-packages (1.0.4)
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Requirement already satisfied: langchain-text-splitters<1.0.0,>=0.3.8 in /usr/local/lib/python3.11/d
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Requirement already satisfied: opentelemetry-exporter-otlp-proto-grpc>=1.2.0 in /usr/local/lib/pythc
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Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.11/dist-packages (from ai
```

Wikipedia Retriever

```
from langchain_community.retrievers import WikipediaRetriever
```

```
# Initialize the retriever (optional: set language and top_k)
retriever = WikipediaRetriever(top_k_results=2, lang="en")
```

```
# Define your query
query = "the geopolitical history of india and pakistan from the perspective of a chinese"
```

```
# Get relevant Wikipedia documents
docs = retriever.invoke(query)
```

```
docs
```

```
[Document(metadata={'title': 'United States aid to Pakistan', 'summary': 'The United States had been providing military aid and economic assistance to Pakistan for various purposes since 1948. Recently U.S. stopped military aid to Pakistan, which was about US$2 billion per year. With U.S. military assistance suspended in 2018 and civilian aid reduced to about $300 million for 2022, Pakistani authorities have turned to other countries for help.', 'source': 'https://en.wikipedia.org/wiki/United_States_aid_to_Pakistan'}, page_content="The United States had been providing military aid and economic assistance to Pakistan for various purposes since 1948. Recently U.S. stopped military aid to Pakistan, which was about US$2 billion per year. With U.S. military assistance suspended in 2018 and civilian aid reduced to about $300 million for 2022, Pakistani authorities have turned to other countries for help.\n\n\n== History ==\nFrom 1947 to 1958, under civilian leadership, the United States provided Pakistan with modest economic aid and limited military assistance. During this period, Pakistan became a member of the South East Asian Treaty Organization (SEATO) and the Central Treaty Organization (CENTO), after a Mutual Defence Assistance Agreement signed in May 1954, which facilitated increased levels of both economic and military aid from the U.S.\nIn 1958, Ayub Khan led Pakistan's first military coup, becoming Chief Martial Law Administrator (CMLA) and later President until 1969. During his tenure, the U.S. delivered substantial economic and military aid, despite Pakistan's governance by military regime, and amidst events like the Indo-Pakistani war of 1965.\nYahya Khan succeeded Ayub in 1969, holding power during the Indo-Pakistani war of 1971 which led to the secession of East Pakistan and the formation of Bangladesh. Under Yahya, the U.S. provided adequate economic but minimal military aid.\nCivilian governance of Zulfikar Ali Bhutto resumed from 1971 to 1977, during which the U.S. offered modest economic support and withheld military aid as Pakistan finalized its constitution, establishing a parliamentary democracy.\nFollowing another military coup in 1977, Muhammad Zia-ul-Haq led the country. In April 1979, President Jimmy Carter halted all aid, excluding food assistance, due to Pakistan's efforts to establish a uranium enrichment facility, following Symington Amendment. Initial U.S. aid was limited, increasing significantly after geopolitical shifts such as the Soviet invasion of Afghanistan in 1979 and the fall of the Shah of Iran. U.S. sanctions imposed in April 1979 due to Pakistan's nuclear activities were lifted by the end of the year in light of these events.\nBetween 1988 and 1999, under civilian and democratic governments, U.S. aid was low, particularly after the Soviet withdrawal from Afghanistan in 1989. The aid was suspended in the 1990s under President George H. W. Bush, who cited concerns over Pakistan's developing nuclear program. Relations between the two countries deteriorated as the U.S. implemented the Pressler Amendment. This amendment led to severe sanctions against Pakistan, exacerbating economic challenges for the country's nascent civilian government. Consequently, all forms of bilateral aid from the U.S. to Pakistan were halted. The once expansive operations of the U.S. Agency for International Development (USAID) in Pakistan, which had employed over 1,000 staff across the country, were dramatically reduced almost overnight. Further complications arose from U.S. sanctions following Pakistan's nuclear tests in 1998, which were conducted in response to similar tests by India.\nThe return of military rule under Pervez Musharraf from 1999 to 2008 initially saw little U.S. economic or military aid. However, following the September 11, 2001 attacks, all U.S. sanctions were removed, and Pakistan, having aligned with the U.S. in the War on Terror, received substantial increases in both economic and military assistance.\nOn June 16, 2009, the U.S. Senate Foreign Relations Committee passed the Enhanced Partnership with Pakistan Act of 2009, commonly referred to as the Kerry-Lugar Bill. The bipartisan act authorized an annual provision of $1.5 billion in U.S. aid to Pakistan, aimed at fostering enhanced bilateral relations.\nIn 2011, the Obama administration suspended more than one-third of all military assistance, totaling approximately $800 million due to Osama bin Laden-related controversy. This reduction encompassed funds designated for military hardw"),
 Document(metadata={'title': 'Foreign relations of Pakistan', 'summary': "The Islamic Republic of Pakistan emerged as an independent country through the partition of India in August 1947 and was admitted as a United Nations member state in September 1947. It is currently the second-largest country within the Muslim world in terms of population, and is also the only Muslim-majority country in possession of nuclear weapons. De facto, the country shares direct land borders with India, Iran, Afghanistan, and China.\nThe country has extensive trade relations with the European Union and with several countries globally. As of 2023, Pakistan does not recognize two other United Nations member states (Armenia and Israel) and its ties with India remain frozen since 2019.\nFrom a geopolitical perspective, Pakistan's location is strategically important as it is situated at the crossroads of major maritime and land transit routes between the Middle East and South Asia, while
```

```
# Print retrieved content
for i, doc in enumerate(docs):
    print(f"\n--- Result {i+1} ---")
    print(f"Content:\n{doc.page_content}...")  # truncate for display
```

```
--- Result 1 ---
Content:
The United States had been providing military aid and economic assistance to Pakistan for various pur


== History ==
From 1947 to 1958, under civilian leadership, the United States provided Pakistan with modest economi
In 1958, Ayub Khan led Pakistan's first military coup, becoming Chief Martial Law Administrator (CMLA
Yahya Khan succeeded Ayub in 1969, holding power during the Indo-Pakistani war of 1971 which led to t
```

```
Civilian governance of Zulfikar Ali Bhutto resumed from 1971 to 1977, during which the U.S. offered m
Following another military coup in 1977, Muhammad Zia-ul-Haq led the country. In April 1979, Presiden
Between 1988 and 1999, under civilian and democratic governments, U.S. aid was low, particularly afte
The return of military rule under Pervez Musharraf from 1999 to 2008 initially saw little U.S. econom
On June 16, 2009, the U.S. Senate Foreign Relations Committee passed the Enhanced Partnership with Pa
In 2011, the Obama administration suspended more than one-third of all military assistance, totaling

--- Result 2 ---
Content:
The Islamic Republic of Pakistan emerged as an independent country through the partition of India in 
The country has extensive trade relations with the European Union and with several countries globally
From a geopolitical perspective, Pakistan's location is strategically important as it is situated at


== Foreign policy of Pakistan ==
Pakistan's foreign policy seeks to 'promote the internationally recognized norms of interstate relati
Pakistan's foreign policy is meant to formalize and define its interactions with foreign nations and


=== M A Jinnah's Vision ===
In 1947, Muhammad Ali Jinnah, founder of the state of Pakistan, clearly described the principles and
```

Vector Store Retriever

```
from langchain_community.vectorstores import Chroma
from langchain_openai import OpenAIEmbeddings
from langchain_core.documents import Document
```

```
# Step 1: Your source documents
documents = [
    Document(page_content="LangChain helps developers build LLM applications easily."),
    Document(page_content="Chroma is a vector database optimized for LLM-based search."),
    Document(page_content="Embeddings convert text into high-dimensional vectors."),
    Document(page_content="OpenAI provides powerful embedding models."),
]
```

```
# Step 2: Initialize embedding model
embedding_model = OpenAIEmbeddings()

# Step 3: Create Chroma vector store in memory
vectorstore = Chroma.from_documents(
    documents=documents,
    embedding=embedding_model,
    collection_name="my_collection"
)
```

```
# Step 4: Convert vectorstore into a retriever
retriever = vectorstore.as_retriever(search_kwargs={"k": 2})
```

```
query = "What is Chroma used for?"
results = retriever.invoke(query)
```

```
for i, doc in enumerate(results):
    print(f"\n--- Result {i+1} ---")
    print(doc.page_content)
```

```
--- Result 1 ---
Chroma is a vector database optimized for LLM-based search.

--- Result 2 ---
LangChain helps developers build LLM applications easily.
```

```
results = vectorstore.similarity_search(query, k=2)
```

```
for i, doc in enumerate(results):
    print(f"\n--- Result {i+1} ---")
    print(doc.page_content)
```

```
--- Result 1 ---
Chroma is a vector database optimized for LLM-based search.

--- Result 2 ---
LangChain helps developers build LLM applications easily.
```

MMR

```
# Sample documents
docs = [
    Document(page_content="LangChain makes it easy to work with LLMs."),
    Document(page_content="LangChain is used to build LLM based applications."),
    Document(page_content="Chroma is used to store and search document embeddings."),
    Document(page_content="Embeddings are vector representations of text."),
    Document(page_content="MMR helps you get diverse results when doing similarity search."),
    Document(page_content="LangChain supports Chroma, FAISS, Pinecone, and more."),
]
```

```
from langchain_community.vectorstores import FAISS

# Initialize OpenAI embeddings
embedding_model = OpenAIEmbeddings()

# Step 2: Create the FAISS vector store from documents
vectorstore = FAISS.from_documents(
    documents=docs,
    embedding=embedding_model
)
```

```
# Enable MMR in the retriever
retriever = vectorstore.as_retriever(
    search_type="mmr",                   # <-- This enables MMR
    search_kwargs={"k": 3, "lambda_mult": 0.5}  # k = top results, lambda_mult = relevance-diversity
)
```

```
query = "What is langchain?"
results = retriever.invoke(query)
```

```
for i, doc in enumerate(results):
    print(f"\n--- Result {i+1} ---")
    print(doc.page_content)
```

```
--- Result 1 ---
LangChain is used to build LLM based applications.

--- Result 2 ---
Embeddings are vector representations of text.

--- Result 3 ---
LangChain supports Chroma, FAISS, Pinecone, and more.
```

Multiquery Retriever

```
from langchain_community.vectorstores import FAISS
from langchain_openai import OpenAIEmbeddings
from langchain_core.documents import Document
from langchain_openai import ChatOpenAI
from langchain.retrievers.multi_query import MultiQueryRetriever
```

```
# Relevant health & wellness documents
all_docs = [
    Document(page_content="Regular walking boosts heart health and can reduce symptoms of depression
    Document(page_content="Consuming leafy greens and fruits helps detox the body and improve longev
    Document(page_content="Deep sleep is crucial for cellular repair and emotional regulation.", met
    Document(page_content="Mindfulness and controlled breathing lower cortisol and improve mental cl
    Document(page_content="Drinking sufficient water throughout the day helps maintain metabolism an
```

```
    Document(page_content="The solar energy system in modern homes helps balance electricity demand.
    Document(page_content="Python balances readability with power, making it a popular system design
    Document(page_content="Photosynthesis enables plants to produce energy by converting sunlight.",
    Document(page_content="The 2022 FIFA World Cup was held in Qatar and drew global energy and exci
    Document(page_content="Black holes bend spacetime and store immense gravitational energy.", meta
]
```

```
# Initialize OpenAI embeddings
embedding_model = OpenAIEmbeddings()

# Create FAISS vector store
vectorstore = FAISS.from_documents(documents=all_docs, embedding=embedding_model)
```

```
# Create retrievers
similarity_retriever = vectorstore.as_retriever(search_type="similarity", search_kwargs={"k": 5})
```

```
multiquery_retriever = MultiQueryRetriever.from_llm(
    retriever=vectorstore.as_retriever(search_kwargs={"k": 5}),
    llm=ChatOpenAI(model="gpt-3.5-turbo")
)
```

```
# Query
query = "How to improve energy levels and maintain balance?"
```

```
# Retrieve results
similarity_results = similarity_retriever.invoke(query)
multiquery_results= multiquery_retriever.invoke(query)
```

```
for i, doc in enumerate(similarity_results):
    print(f"\n--- Result {i+1} ---")
    print(doc.page_content)

print("*"*150)

for i, doc in enumerate(multiquery_results):
    print(f"\n--- Result {i+1} ---")
    print(doc.page_content)
```

```
--- Result 1 ---
Drinking sufficient water throughout the day helps maintain metabolism and energy.

--- Result 2 ---
Mindfulness and controlled breathing lower cortisol and improve mental clarity.

--- Result 3 ---
Regular walking boosts heart health and can reduce symptoms of depression.

--- Result 4 ---
Deep sleep is crucial for cellular repair and emotional regulation.

--- Result 5 ---
The solar energy system in modern homes helps balance electricity demand.
****************************************************************************************************************************************

--- Result 1 ---
Drinking sufficient water throughout the day helps maintain metabolism and energy.

--- Result 2 ---
Mindfulness and controlled breathing lower cortisol and improve mental clarity.

--- Result 3 ---
Regular walking boosts heart health and can reduce symptoms of depression.

--- Result 4 ---
Consuming leafy greens and fruits helps detox the body and improve longevity.

--- Result 5 ---
Deep sleep is crucial for cellular repair and emotional regulation.
```

ContextualCompressionRetriever

```
from langchain_community.vectorstores import FAISS
from langchain_openai import OpenAIEmbeddings, ChatOpenAI
from langchain.retrievers.contextual_compression import ContextualCompressionRetriever
from langchain.retrievers.document_compressors import LLMChainExtractor
from langchain_core.documents import Document
```

```
# Recreate the document objects from the previous data
docs = [
    Document(page_content=(
        """The Grand Canyon is one of the most visited natural wonders in the world.
        Photosynthesis is the process by which green plants convert sunlight into energy.
        Millions of tourists travel to see it every year. The rocks date back millions of years."""
    ), metadata={"source": "Doc1"}),

    Document(page_content=(
        """In medieval Europe, castles were built primarily for defense.
        The chlorophyll in plant cells captures sunlight during photosynthesis.
        Knights wore armor made of metal. Siege weapons were often used to breach castle walls."""
    ), metadata={"source": "Doc2"}),

    Document(page_content=(
        """Basketball was invented by Dr. James Naismith in the late 19th century.
        It was originally played with a soccer ball and peach baskets. NBA is now a global league.""
    ), metadata={"source": "Doc3"}),

    Document(page_content=(
        """The history of cinema began in the late 1800s. Silent films were the earliest form.
        Thomas Edison was among the pioneers. Photosynthesis does not occur in animal cells.
        Modern filmmaking involves complex CGI and sound design."""
    ), metadata={"source": "Doc4"})
]
```

```
# Create a FAISS vector store from the documents
embedding_model = OpenAIEmbeddings()
vectorstore = FAISS.from_documents(docs, embedding_model)
```

```
base_retriever = vectorstore.as_retriever(search_kwargs={"k": 5})
```

```
# Set up the compressor using an LLM
llm = ChatOpenAI(model="gpt-3.5-turbo")
compressor = LLMChainExtractor.from_llm(llm)
```

```
# Create the contextual compression retriever
compression_retriever = ContextualCompressionRetriever(
    base_retriever=base_retriever,
    base_compressor=compressor
)
```

```
# Query the retriever
query = "What is photosynthesis?"
compressed_results = compression_retriever.invoke(query)
```

```
for i, doc in enumerate(compressed_results):
    print(f"\n--- Result {i+1} ---")
    print(doc.page_content)
```

```
--- Result 1 ---
Photosynthesis is the process by which green plants convert sunlight into energy.

--- Result 2 ---
The chlorophyll in plant cells captures sunlight during photosynthesis.
```